

Adapting Falcon3-7B Language Model for Arabic: Methods, Challenges, and Outcomes

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Abstract

Under-represented languages suffer from a lack of data, and as a result, there are few LLMs that support them. Extending an existing LLM to a new language is a practical option for startups, university labs, and organizations with limited budgets. This process involves several steps. In this paper, we describe how we adapted the Falcon3-7B model to Arabic, covering everything from data collection and training to evaluation. Falcon-Arabic was trained exclusively on native data to better capture the cultural and linguistic aspects of the language. Our evaluations show that Falcon-Arabic achieves state-of-the-art results on a range of Arabic benchmarks.

1 Introduction

Despite the rapid advancement of Large Language Models (LLMs) across many languages, Arabic remains comparatively underserved. Most state-of-the-art multilingual models such as mBERT (Devlin et al., 2018), XLM-R (Conneau et al., 2020), and BLOOM (Workshop et al., 2023) have enabled substantial cross-lingual transfer, but Arabic presents unique linguistic features that introduce specific challenges requiring tailored modeling strategies (Abdul-Mageed et al., 2021). The morphological richness of Arabic, with its complex root-and-pattern system, extensive inflection, and agglutinative properties, poses significant challenges for tokenization and representation (Habash et al., 2013).

The scarcity of high-quality Arabic data exacerbates these challenges, compelling researchers to rely heavily on machine translation to augment dataset sizes (Koehn, 2005). However, this approach often results in models misaligned with Arabic cultural contexts and linguistic nuances (Bender et al., 2021). As demonstrated in Figure 1, when asked to explain an Arabic proverb, several prominent LLMs failed to grasp the underlying cultural meaning, instead providing literal word-by-word

Prompt and Responses from Multiple LLMs
Prompt: ما معنى "يَدُّك في الموية الباردة" في السياق الاجتماعي؟
Qwen2.5-7B-Instruct: العبارة "يَدُّك في الموية الباردة" هي عبارة شائعة في بعض الثقافات العربية، وتستخدم عادة في السياق الاجتماعي للإشارة إلى حالة من الارتباك أو القوضى أو عدم الراحة.
Llama-3.1-8B-Instruct: في السياق الاجتماعي، يعني "يَدُّك في الموية الباردة" أنك محتاجون للمساعدة، خاصة إذا كنت في وضع صعب أو تعاني من صعوبات مالية.
Falcon3-7B-Arabic: يَدُّك في الموية الباردة هي عبارة شعبية تستخدم في السياق الاجتماعي للإشارة إلى شخص يعيش في حالة من الراحة أو الرفاهية، حيث لا يواجه أي تحديات أو صعوبات.

Figure 1: Example of LLM responses to an Arabic prompt with cultural and dialectal elements; Falcon-Arabic correctly captured the cultural nuance. The English translation of the example is provided in the Appendix (Figure 4).

translations. This limitation underscores the need for culturally-aware Arabic language models that can capture the depth and subtlety of Arabic expression.

Arabic LLMs can be categorized into three main model families: native models trained from scratch, multilingual models with Arabic support, and models adapted from existing multilingual LLMs (Mashaabi et al., 2024). Training Arabic models from scratch requires trillions of Arabic tokens, which are difficult to collect, along with substantial computational infrastructure (Kaplan et al., 2020).

Analysis of the Open Arabic LLM Leaderboard (El Filali et al., 2025) reveals that multilingual models such as Qwen (Qwen et al., 2025) and LLaMA (Grattafiori et al., 2024), as well as adapted models like AceGPT (Huang et al., 2024) consis-

tently rank among the top performers. Adapting existing LLMs to new languages requires significantly less data and computational resources compared to training from scratch (Wang et al., 2025). The foundation model already possesses general knowledge, reasoning capabilities, and common sense, making it a matter of aligning new language tokens with existing representations rather than learning from scratch. This approach has proven successful in recent continual pretraining studies (Gupta et al., 2023).

Motivated by these findings, we adapt Falcon3-7B (Team, 2024a) to Arabic. The adaptation process presents unique challenges since Falcon3-7B’s tokenizer lacks Arabic support, requiring careful vocabulary extension and embedding initialization (Minixhofer et al., 2022). In this work, we detail the complete adaptation pipeline, from data collection and tokenizer extension to model layer adaptation, multi-stage training, and post-training procedures. We document the challenges encountered and key insights gained, contributing valuable knowledge to the community for future language adaptation efforts.

What distinguishes Falcon-Arabic is our exclusive use of native Arabic datasets without machine translation, encompassing diverse content including dialects, poetry, literature, and contemporary texts, all authentically Arabic. Through training on only 600B tokens, we achieve a model that outperforms LLMs two times its size while maintaining strong cultural relevance and linguistic authenticity for the Arabic-speaking community. Our approach demonstrates that targeted adaptation with high-quality, culturally-authentic data can achieve superior performance compared to larger, more resource-intensive alternatives (Touvron et al., 2023).

2 Related Work

The interest in building Arabic Language Models has emerged with multiple initiatives spanning various sizes from a few million parameters to billions (Mashaabi et al., 2024). Models like AraBERT (Abdul-Mageed et al., 2021) and AraGPT2 (Antoun et al., 2021) were among the first transformer-based Arabic LLMs with millions of parameters (Vaswani et al., 2017). AraBERT introduced comprehensive pre-training on Arabic text with careful preprocessing to handle the language’s morphological complexity and diacritization variations. AraGPT2 demonstrated the effectiveness of gen-

erative pre-training for Arabic text generation, establishing foundational benchmarks for subsequent Arabic language models. Subsequently, increasing the number of parameters in these models showed promising performance improvements, leading to more ambitious initiatives toward building Arabic Large Language Models. Arabic LLMs can be categorized into three main categories based on how Arabic was incorporated into the training data.

Native Arabic Models are trained on Arabic from scratch or with Arabic as a primary language. JAIS (Sengupta et al., 2023) represents a prominent example of this category, being trained on a balanced mix of Arabic, English, and code to achieve strong performance across Arabic dialects while maintaining multilingual capabilities. The model was specifically designed to handle the nuances of Arabic script and cultural context. Other small Arabic LLMs trained from scratch include ArabianGPT (Koubaa et al., 2024) and AraGPT (Antoun et al., 2021).

Multilingual Foundation Models constitute the second category, typically featuring strong English support as a primary language while demonstrating competitive results across other languages, including Arabic. The LLaMA family of models (Grattafiori et al., 2024) supports a wide range of languages through extensive multilingual pre-training, showing robust cross-lingual transfer capabilities. Qwen2.5 (Qwen et al., 2025) and Qwen3 (Yang et al., 2025) have demonstrated strong multilingual performance with particular attention to maintaining quality across diverse writing systems. The Gemma (Team et al., 2024a) and Gemma 2 (Team et al., 2024b) models have shown promising results in multilingual settings while maintaining computational efficiency through architectural innovations.

Adapted Arabic Models represent the third category, comprising models that were fine-tuned or adapted from multilingual LLMs to enhance Arabic-specific performance. Some models were adapted from LLaMA such as AceGPT (Huang et al., 2024), JAIS adapted family (Sengupta et al., 2023), Yehia (Navid-AI, 2025). While others were adapted from Gemma such as SILMA (Team, 2024b) and Fanar (Team et al., 2025). Each model targets specific improvements: AceGPT focuses on cultural adaptation, ALLAM emphasizes Arabic linguistic features, while Yehia and Fanar enhance regional dialect support. The JAIS adapted family and SILMA demonstrate continued progress in instruction fol-

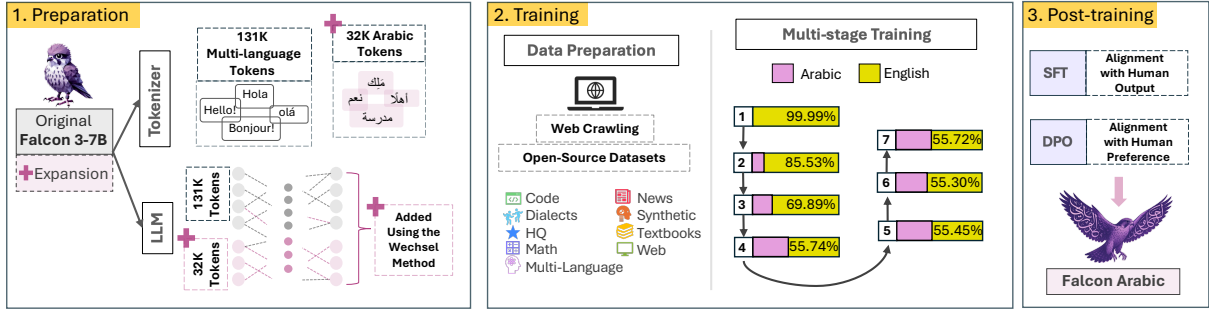


Figure 2: Schematic View of the adaptation of Falcon-3 7B Language Model for Arabic.

lowing and conversational capabilities for Arabic.

While these LLMs demonstrate competitive performance across multiple benchmarks, multilingual models such as Gemma, LLaMA, and Qwen often lack culturally-centric data related to Arabic and the Arab region, heavily relying on machine translation which may introduce cultural and linguistic biases. To address these limitations, we built Falcon-Arabic by training exclusively on native Arabic data and carefully designing training stages to smoothly integrate culturally and linguistically relevant content into Falcon3-7B, ensuring authentic representation of Arabic language nuances and cultural contexts.

3 Datasets

Addressing the significant challenge of limited Arabic data availability, we prepared a comprehensive multilingual corpus totaling approximately 600 BT tokens, with Arabic comprising 40% and English 60% of the dataset.

3.1 Arabic Datasets

Recognizing the crucial gap in Arabic datasets for LLMs, particularly in dialectal diversity and STEM-related content, we developed a comprehensive Arabic corpus addressing these limitations. The dataset covers diverse Arabic dialects including Levantine (الشام), Maghrebi (Darija), Egyptian and Gulf Arabic, ensuring broad linguistic representation across multiple textual domains: web documents (Penedo et al., 2025), educational materials, news sources, and mathematical content.

For low-resource dialects, we leveraged recent Moroccan Darija adaptations (Shang et al., 2024) and specialized OCR datasets from Arabic-Nougat (Rashad, 2024). Additionally, we actively crawled and curated new data from educational books, web documents, and news articles. A distinctive feature is our focus on grammatical details, including annotations for grammatical structures (إعراب) and

various linguistic forms. Critically, we avoided machine-translated content, instead selecting authentic Arabic language data from different historical periods to maintain performance quality.

3.2 English Datasets

Acknowledging the importance of maintaining robust English performance alongside Arabic proficiency, a comprehensive English corpus comprising approximately 60% of the total dataset was curated. This dataset covers diverse textual domains including extensive collections from textbooks, web sources (Penedo et al., 2025; Lozhkov et al., 2024a; Ben Allal et al., 2024), synthetic data, code repositories (Lozhkov et al., 2024b), high-quality documents, mathematical texts (Han et al., 2024), and multilingual content. While the dataset was not fully expanded from prior training data, it strategically combines previously effective resources with newly introduced data with the aim of enhancing performance across key benchmarks.

To ensure balanced representation and address domain gaps, we further supplemented the dataset with synthetically generated data and additional crawled resources, including recent news articles and educational materials.

4 Approach

In this section we detail the steps that we followed to adapt Falcon3-7B to Arabic.

4.1 Tokenizer Extension

The original Falcon3-7B tokenizer primarily covers English, French, Spanish, and Portuguese, making it inefficient for Arabic text due to over-segmentation. To address this, we extend Falcon’s vocabulary by adding 32,768 Arabic tokens to the original 131,072 tokens, resulting in a total vocabulary of 163,840 tokens which remains a reasonable tokenizer size for a 7B LLM.

Model	Fertility Score	Vocabulary Size
Falcon-Arabic	2.17	163,840
Gemma-3-4B	2.18	262,208
Llama-3.1-8B	2.43	128,256
Qwen2.5-7B	2.55	152,064
Falcon3-7B-Base	4.54	131,072

Table 1: Fertility scores of different LLMs. Lower is better.

We trained a BPE tokenizer on the Arabic subset of FineWeb2 (Penedo et al., 2025) using the same configuration as Falcon3-7B, then merged the vocabularies while preserving original token mappings. We evaluated the effectiveness by computing fertility scores¹ (average tokens per word) for both tokenizers on Arabic text, with results shown in Table 1.

This extension provides reduced training and inference costs, lower latency, and support for longer context windows (Gosal et al., 2024). Models with low fertility tokenizers demonstrate improved performance on downstream tasks (Ahuja et al., 2023).

4.2 Layers Extension

After training a new Arabic tokenizer and extending the Falcon3-7B tokenizer, we needed to incorporate the newly added tokens into both the input embedding layer and the output layer (lm head). The critical challenge lies in properly initializing the embeddings associated with these new tokens to maintain model performance and training stability. Multiple initialization approaches exist for newly added token embeddings, including zero, random, and averaging existing embeddings (de Vries and Nissim, 2021; Marchisio et al., 2023; Zhao et al., 2024). However, according to Gosal et al. (2024), these conventional approaches may lead to degraded performance as they deviate from the initial distribution of pre-trained word embeddings.

To address these limitations, we apply the Wechsel approach (Minixhofer et al., 2022) to initialize the newly added token embeddings. This method leverages cross-lingual alignment and subword-level correspondences to create more informed initializations that preserve the semantic structure of the original embedding space.

The Wechsel method proceeds through the following key steps: (1) tokenize bilingual dictionary

words into subwords using both tokenizers, (2) compute subword embeddings e_{sw} using fastText (Bojanowski et al., 2016) as the sum of n-gram embeddings $N(sw)$ as in Equation 1 (3) align subword embeddings across languages using Orthogonal Procrustes alignment (Schönemann, 1966; Artetxe et al., 2016), (4) initialize new token embeddings e_{sw_t} as weighted averages of source embeddings using cosine similarity as weights Equation 2, and (5) copy non-embedding parameters from the source model.

$$e_{sw} = \sum_{ng \in N(sw)} e_{ng} \quad (1)$$

$$e_{sw_t} = \frac{\sum_{sw_s \in N(sw_t)} \text{sim}(sw_s, sw_t) \cdot e_{sw_s}}{\sum_{sw_s \in N(sw_t)} \text{sim}(sw_s, sw_t)} \quad (2)$$

where e_{sw} is the embedding of subword sw , $N(sw)$ is the set of n-grams occurring in the subword, e_{ng} is the embedding of n-gram ng , e_{sw_t} is the target subword embedding, $N(sw_t)$ represents the set of neighboring subwords in the source language, and $\text{sim}(sw_s, sw_t)$ denotes the cosine similarity between source and target subwords.

This approach ensures that newly added Arabic tokens receive semantically meaningful initializations that are consistent with the pre-trained embedding space, thereby facilitating more efficient adaptation and improved performance on Arabic language tasks.

4.3 Continuous Pretraining

With the tokenizer extended and the input and output embedding layers properly initialized, the model is ready for continuous pretraining. We designed a multi-stage training approach consisting of four stages to carefully control the data mixture, sequence length, and ratio between Arabic and English content. Table 2 summarizes the percentage of each data source per stage and the corresponding sequence lengths used.

The first stage represents the longest training phase with the shortest sequence length, as most datasets contain relatively short sequences. This approach is more computationally efficient and requires fewer resources while maintaining training stability. Stages 2 and 3 are designed to extend the context length capabilities of Falcon-Arabic to 16K and 32K tokens, respectively. We conclude the pre-training with a decay stage to stabilize convergence

¹Dataset used from <https://huggingface.co/spaces/wissamantoun/arabic-tokenizers-leaderboard>

Stage	Seq length	Textbooks	Code	HQ	Math	Synthetic	Dialects	News	Multilang	Web
1.1	8K	11.74	13.85	14.69	2.94	15.67	0.00	0.00	0.58	40.53
1.2	8K	0.69	3.69	29.13	15.55	0.00	0.00	7.54	0.66	42.74
1.3	8K	1.72	5.23	9.97	8.20	15.31	0.11	0.46	0.83	58.17
1.4	8K	11.65	11.93	3.36	12.08	5.54	0.06	1.77	0.46	53.15
2	16K	31.59	9.71	13.51	4.74	5.89	0.13	3.38	1.27	29.78
3	32K	38.58	2.71	3.23	16.08	15.70	0.17	0.29	0.38	22.86
Decay	32K	18.89	1.61	4.85	30.25	12.56	0.16	22.08	0.20	9.40

Table 2: Training stages of Falcon-Arabic.

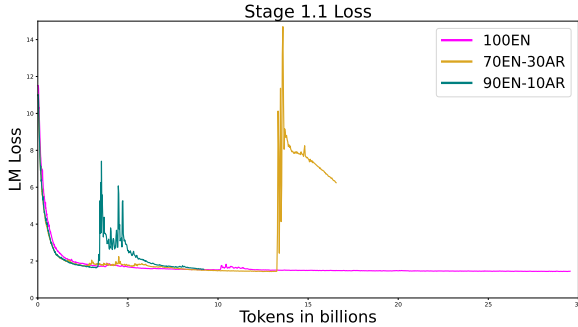


Figure 3: Training loss.

and prevent overfitting as the model approaches optimal performance. This final stage employs learning rate decay to enable smaller, more precise parameter updates, allowing the model to fine-tune its internal representations without overshooting minima or introducing instability.

Since Falcon3-7B was not originally exposed to Arabic data during its pretraining, introducing Arabic datasets requires careful consideration to avoid catastrophic forgetting and important distribution shifts (Çagatay Yildiz et al., 2024). We conducted multiple experiments for the first training stage to identify the optimal proportion of Arabic data while monitoring training loss stability. As shown in Figure 3, initiating training with 30% Arabic data resulted in significant training instability, evidenced by substantial loss spikes. Reducing the Arabic percentage to 10% improved stability but still exhibited spikes, suggesting the model required additional English data for stabilization.

To address this challenge, we implemented a short stabilization stage of 29BT consisting of 100% English data, allowing the model to adjust to the newly added tokens gradually. Following this adjustment period, we employed three additional sub-stages where we progressively increased the Arabic data percentage to achieve 45%, which we maintained across the remaining training stages as detailed in Table 3. This gradual approach ensures

Stage	Arabic	Other Languages	Total
1.1	0.00	29.55	29.55
1.2	5.54	32.74	38.28
1.3	13.83	32.09	45.92
1.4	78.30	98.61	176.91
2	38.61	48.06	86.67
3	28.62	34.00	62.62
Decay	57.39	69.34	126.73

Table 3: Distribution of Arabic and other Languages in Billion Tokens (BT) at each training stage.

smooth integration of Arabic content while preserving the model’s existing capabilities and maintaining training stability throughout the continuous pretraining process.

Checkpoints of each training stage were evaluated separately on Arabic and English benchmarks to monitor the evolution of the training process and detect early signs of catastrophic forgetting or bad data. More details are provided in Section 6. Falcon-Arabic was trained on 566B tokens using 32 H100 nodes (8k toks/GPU/s), corresponding to 3.4 days of wall-clock training and 2.5×10^{22} FLOPs.

5 Post-training

At this stage, we trained our base model to engage in conversations and follow user instructions. We employed Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) to obtain an instruct version of the Falcon-Arabic.

5.1 Supervised Fine-tuning (SFT)

We started by performing SFT, to make the model capable of conducting conversations, making it capable of following instructions and answering questions. In line with continuous pretraining, both arabic and english data were fed to the model at this stage, ensuring that it could chat in both languages. Next, we discuss the SFT datasets used.

Hyperparameter	SFT	DPO
Batch Size	512	128
Epochs	3	1
KL Penalty (B)	-	5
<i>Optimizer</i>		
Optimizer	AdamW	AdamW
B_1	0.9	0.9
B_2	0.99	0.99
ϵ	1×10^{-15}	1×10^{-8}
<i>Learning Rate</i>		
Decay Type	linear	linear
Max lr	1×10^{-6}	1×10^{-7}
Min lr	6×10^{-8}	-
Warmup	3%	5%

Table 4: SFT/DPO Optimal Hyperparameters.

5.1.1 SFT Data

A wide range of datasets was used in terms of category and task type, curated from public datasets and curated sources. Examples of Arabic public datasets used are Aya (Singh et al., 2024), WikiReading (Albilali et al., 2022), and Bactrian-X (Li et al., 2023). Furthermore, an in-house synthetic SFT dataset was created that expands the list of covered topics and allows the model to handle multi-turn chats. To ensure the chat model remains multilingual, the publicly available tulu-3 dataset (Lambert et al., 2025) was used. The resulting SFT dataset comprised 4.3 million samples, with a language distribution of approximately 55% Arabic and 45% English.

5.1.2 SFT Recipe

An extensive search was performed on the SFT hyperparameters to select the optimal set of hyperparameters values that maximizes the model performance. Table 4 shows the optimal SFT configuration we used during the SFT stage.

5.2 Direct Preference Optimization (DPO)

In the second stage of the post-training, we leveraged DPO (Rafailov et al., 2024) to align the model with generating more human-like responses. DPO offered an offline training approach, where the need for a reward model is alleviated. Typically, DPO is applied to binary preference data, where each sample has a pair of accepted and rejected responses for the same prompt. The ultimate objective of this stage is to steer the model to become aligned with human preference while maintaining its knowledge and capabilities from the SFT stage. Several public binary preference datasets were utilized, such as

argilla², orca³, and tulu-3 (Lambert et al., 2025). The optimal hyperparameters found for DPO is shown in Table 4.

6 Evaluation

To assess the performance of Falcon-Arabic, the pre-trained and instruct models were evaluated using several benchmarks⁴. The backend of our evaluation setup leveraged lighteval (Habib et al., 2023) and lm-eval (Gao et al., 2024), which are both established evaluation tools within the NLP community. We compared Falcon-Arabic against several open-source SOTA models ($< 14B$), chosen based on the OALL (El Filali et al., 2025). The benchmarks used in this work are discussed in the following subsections.

6.1 Benchmarks

General benchmarks AlGhafa (Almazrouei et al., 2023) is an Arabic benchmark that targets the evaluation of tasks that include comprehension, sentiment analysis, and question-answering. Only the native Arabic datasets were used. ArabicMMLU is a native Arabic benchmark, which includes 40 tasks and nearly 15k MCQs (Koto et al., 2024). ArbMMLU-HT is a human translated version of the original English MMLU dataset containing 57 tasks. Subjects covered in ArabicMMLU and ArabicMMLU-HT span various topics such as history and social science, which are of varying complexity (Sengupta et al., 2023). Exams (Hardalov et al., 2020) is a benchmark of questions that targets high school level of difficulty, and only the Arabic samples were used. MadinahQA (Koto et al., 2024) is a benchmark with 983 QA pairs that focuses generally on the syntax and grammar of the Arabic language.

Reasoning To access the reasoning capabilities of our model, we integrated the publicly available dataset, called Arabic-GSM8K⁵ with lighteval, which is a translation of the GSM8K (Cobbe et al., 2021a).

RAG ALRAGE (El Filali et al., 2025) is a benchmark composed of 2.1k QA pairs that were generated from 40 Arabic books. ALRAGE is intended for the evaluation of LLMs’ retrieval-augmented generation (RAG) capabilities in Arabic. The tasks

²2A2I/argilla-dpo-mix-7k-arabic

³multilingual/orca_dpo_pairs

⁴chat-template was used for Instruct models

⁵<https://huggingface.co/datasets/Omartificial-Intelligence-Space/Arabic-GSM8K>

Model	Size	ALGhafa	ArabicMMLU	EXAMS	MadinahQA	AraTrust	ALRAGE	ArbMMLU-HT	Avg
Qwen2.5	7B	72.17	61.42	49.16	51.13	77.56	64.83	51.67	61.13
jais-adapted	7B	32.92	27.33	26.44	24.84	33.91	41.43	27.4	30.61
	13B	40.62	36.97	34.26	29.04	61.18	62.53	33.12	42.53
AceGPT-v2	8B	46.32	50.41	43.58	40.81	69.25	57.76	35.62	49.12
AceGPT	13B	48.23	41.38	36.87	35.37	56.51	79.96	32.12	47.21
Llama-3.1	8B	64.34	52.28	40.04	43.08	71.98	47.08	42.67	51.64
Falcon3-7B-Base	7B	37.89	31.81	24.77	24.87	49.89	60.23	25.88	36.48
Falcon-Arabic	7B	67.17	64.85	52.89	48.79	85.36	63.71	55.25	62.57

Table 5: Falcon-Arabic compared to the best open source SOTA Models. **Bold** indicates the best score in each column; underline indicates the second best.

in this benchmark include questions and target answers, and candidate context, where outputs are judged by Qwen2.5-72B-Instruct.

Truthfulness AraTrust (Alghamdi et al., 2024) is a benchmark with 522 human written MCQs, with the aim of assessing the safety and truthfulness of a model.

Dialect and Culture ArabCulture (Sadallah et al., 2025) was used to assess arab cultural understanding and awareness with questions spanning countries in the Gulf, Levant, North Africa, and the Nile valley. AraDiCE (Mousi et al., 2024) is benchmark composed of 45k samples that includes the dialects translation of major benchmarks, which are ArabicMMLU, boolQ, truthfulqa, piqa, openbookqa, and winogrande in both the Egyptian and the Levantine dialect. Furthermore, the benchmark includes a range of cultural questions related to several Arab countries. For AraDiCE, we report three scores which are Aradice-CULT, Aradice-LEV and Aradice-EGY that corresponds to the mean scores obtained in cultural questions, Levantine questions, and Egyptian samples, respectively.

English benchmarks Considering that Falcon-Arabic was trained to be a multilingual model, its capabilities were evaluated on english tasks too. Therefore, Falcon-Arabic was benchmarked on the open source LLM leaderboard v1 and v2 tasks, which are GSM8K (Cobbe et al., 2021b), HellaSwag (Zellers et al., 2019), ARC Challenge (Clark et al., 2018), Winogrande (Sakaguchi et al., 2021), TruthfulQA (Lin et al., 2022), MMLU (Hendrycks et al., 2021a), IFEval (Zhou et al., 2023), GPQA (Rein et al., 2023), MMLU-pro (Wang et al., 2024), MATH (Hendrycks et al., 2021b), BBH (Suzgun et al., 2022), and MUSR (Sprague et al., 2024).

The evaluation metric used with most of the mentioned benchmarks is normalized accuracy, with the exception of ALRAGE and Arabic-GSM8K. For ALRAGE, an LLM judge was used specifically Qwen2.5-72B-Instruct, whereas *exact match* was used for Arabic-GSM8K.

6.2 Results and Discussion

In this section, we discuss the evaluation results of Falcon-Arabic and other SOTA models on general Arabic, reasoning, cultural and English benchmarks.

6.2.1 Arabic General Benchmarks

Table 5 presents the scores of the Falcon-Arabic model against SOTA models. From the results, it is evident that Falcon-Arabic significantly outperforms the SOTA models in ArabicMMLU, ArbMMLU-HT, and EXAMS. This indicates that our base model excels in general knowledge and STEM subjects. Similar observations can be made in AraTrust, which suggests that Falcon-Arabic is performing the best in terms of safety. Looking at the Alghafa and MadinahQA benchmarks, our model came second to Qwen2.5-7B. Furthermore, in terms of RAG capabilities, our model ranked third, with clear superiority to AceGPT-13B. By viewing the average column, it can be deduced that Falcon-Arabic is superior to all competitors, as manifested by the highest average score of 62.57.

Next, the evaluation of the instruct models’ scores are depicted in Table 6. In the general knowledge and STEM benchmarks, Falcon-Arabic-Instruct obtained the highest scores in ArabicMMLU and ArbMMLU-HT, and ranked second EXAMS benchmark. Looking at MadinahQA, it can be inferred that Falcon-Arabic-Instruct model excelled in grammar tasks, as it achieved the highest score. Despite not performing the best with AraTrust, our instruct model is still on par with the best instruct models, where Yehia-7B-preview scored the highest.

The same observation can be made with Alghafa, where our instruct model is comparable with the best performing models, namely c4ai-command-r7b-arabic. To compare the overall performances, the average score indicates that the Falcon-Arabic-Instruct is superior to all other SOTA models of similar scale ($< 14B$). By com-

Model	Size	ALGhafa	ArabicMMLU	EXAMS	MadinahQA	AraTrust	ALRAGE	ArbMMLU-HT	Avg
Qwen2.5-Instruct	7B	65.6	52.25	39.66	62.73	80.68	77.37	40.33	59.8
Jais-adapted-chat	7B	63.38	49.9	47.71	34.79	66.02	63.6	37.97	51.05
	13B	67.28	54.23	47.3	44.2	79.68	68.41	45.45	58.08
AceGPT-v2	8B	<u>73.48</u>	61.32	49.72	55.89	74.19	70.94	50.89	62.35
AceGPT	13B	59.18	49.84	40.97	33.08	65.7	<u>79.75</u>	39.31	52.55
Llama-3.1-Instruct	8B	70.91	53.58	50.28	39.72	75.57	49.89	47.94	55.41
c4ai-command-r7b-arabic	7B	74.84	59.34	64.99	<u>63.84</u>	80.47	75.9	50.14	<u>67.07</u>
aya-expanse-8b	8B	66.71	57.55	45.44	48.74	82.54	75.78	49.22	60.85
ALLaM-Instruct-preview	7B	69.49	<u>64.9</u>	51.58	54.24	<u>86.93</u>	76.81	52.81	65.25
Yehia-preview	7B	70.81	<u>64.9</u>	52.14	54.37	87.49	76.64	<u>53.4</u>	65.68
SILMA-Instruct-v1.0	9B	33.99	62.16	51.4	52.48	82.83	80.39	40.32	57.64
Falcon3-Instruct	7B	55.75	41.2	29.42	34.4	57.85	43.21	33.59	42.3
Falcon-Arabic-Instruct	7B	72.37	68.27	<u>53.45</u>	73.63	82.62	72.26	55.47	68.3

Table 6: Falcon-Arabic-Instruct compared to the best open source SOTA instruct models on OALL benchmark. **Bold** indicates the best score in each column; underline indicates the second best.

paring Tables 5 and 6, it can be concluded that Falcon-Arabic-Instruct showed an improvement over Falcon-Arabic in all benchmarks, except with the AraTrust benchmark.

6.2.2 Cultural and Reasoning Benchmarks

Table 7, where scores on cultural knowledge and reasoning benchmarks are presented. Looking at Arabic-GSM8K, our model obtained 54.89 Qwen2.5-Instruct scoring the highest in the range of 62. The columns ArabCulture and Aradice-CULT in Table 7, depict the performance of our model and SOTA in existing cultural benchmarks. In both columns, we see solid performance of Falcon-Arabic-Instruct compared to SOTA, evident by sharing the best score in Aradice-CULT and being only 6 points away from the highest scoring model in ArabCulture. Looking at Table 7, we see that Falcon-Arabic-Instruct obtained comparable scores to high performing models in both Levantine and Egyptian dialects by being approximately 2 points away from the best model.

6.2.3 English Benchmarks

Although our primary goal was Arabic adaptation of Falcon3-7B, maintaining English performance remained crucial. We monitored Falcon-Arabic’s English benchmark performance throughout training (detailed in Section 6.1). Figure 6 reveals minimal English performance gains, likely because our English data overlapped with Falcon3-7B’s original training corpus, providing no additional benefit. Table 8 confirms this observation, showing performance degradation in English capabilities. Future work should focus on incorporating novel, high-quality English data during both training and post-training phases to address this limitation.

In summary, Table 5 shows that Falcon-Arabic outperformed all base models shown by the highest average achieved without any close

competition from other models, making it one of the best base models in Arabic tasks. Table 6 shows that Falcon-Arabic-Instruct outscored all competing SOTA models, with solid performance on STEM subjects, Arabic grammar understanding, and truthfulness. However, scores in ALRAGE, indicated that Falcon-Arabic-Instruct is still lacking in RAG capabilities. Table 7 indicates that our instruct model slightly trails in cultural awareness and reasoning, although the performance gap with the leading model is relatively small.

7 Limitations

As with any Large Language Model, Falcon-Arabic is subject to inherent limitations that users must carefully consider (Ashraf et al., 2025). The model can exhibit hallucination behaviors, generating factually incorrect information or fabricating details that appear plausible but are not grounded in reality (Huang et al., 2025). Additionally, despite our efforts to train on high-quality, culturally-authentic Arabic datasets, Falcon-Arabic may still produce toxic, biased, or unsafe content that could be harmful or offensive to users (Mubarak et al., 2024).

The Arabic adaptation of Falcon3-7B reveals a common trade-off in language-specific fine-tuning: while Arabic capabilities improved, English performance declined slightly, indicating that the current adaptation methodology may not optimally balance multilingual retention with Arabic enhancement.

Furthermore, the model’s performance on Arab culture and Arabic-GSM8K benchmarks highlights domain-specific limitations. The cultural knowledge gaps likely stem from insufficient exposure to diverse regional content during training, limiting representation of varied cultural contexts across Arabic-speaking regions. The mathematical reasoning deficiencies on Arabic-GSM8K reflect a

Model	Size	Arabic-GSM8K	ArabCulture	Aradice-CULT	Aradice-LEV	Aradice-EGY
Qwen2.5-Instruct	7B	62.55	53.27	38.89	43.50	45.00
Jais-adapted-chat	7B	10.16	56.86	35	44.87	46.04
	13B	46.25	71.45	40.56	48.41	49.10
AceGPT-v2	8B	45.87	35.44	47.78	49.9	51.04
Llama-3.1-Instruct	8B	49.58	47.53	37.78	43.38	44.79
c4ai-command-r7b-arabic	7B	60.05	67	45	48.44	48.70
aya-expanse-8b	8B	57.77	50.46	47.22	47.66	50.02
ALLaM-Instruct-preview	7B	52.01	67.49	51.67	53.40	53.26
Yehia-preview	7B	50.04	67.58	<u>51.11</u>	51.81	<u>52.52</u>
SILMA-Instruct-v1.0	7B	33.28	71.6	<u>41.67</u>	<u>52.13</u>	52.30
Falcon-Arabic-Instruct	7B	54.89	65.16	51.67	51.01	51.96

Table 7: Falcon-Arabic-Instruct vs. best open source SOTA instruct models on cultural, dialectal and reasoning benchmarks. **Bold** indicates the best score in each column; underline indicates the second best.

Model	IFEval	GPQA	MMLU-pro	BBH	MUSR	MATH	GSM8K	Hellaswag	ARC Challenge	Winogrande	TruthfulQA	MMLU	Avg
	0-shot	0-shot	5-shot	3-shot	0-shot	4-shot	5-shot	10-shot	25-shot	5-shot	0-shot	5-shot	
Falcon3-7B	33.9	12.8	32.34	31.8	18.1	18.5	76.6	75.54	51.0	71.0	37.3	67.4	43.86
Falcon-Arabic	29.1	8.7	28.9	26.6	7.4	12.8	62.0	73.4	49.7	69.9	31.5	60.1	38.34
Falcon3-7B-Instruct	76.12	8.05	34.3	37.92	21.17	40.86	81.5	78.43	62.6	70.4	55.42	70.5	53.11
Falcon-Arabic-Instruct	57.6	4.5	28.3	28.5	19.4	12.3	67.7	71.4	53.5	68.42	31.5	63.34	42.21

Table 8: Falcon model evaluation scores on English benchmarks.

domain mismatch: our model, trained on native Arabic mathematical discourse, struggles with the translated benchmark’s English-centric reasoning patterns and problem formulations that don’t align with authentic Arabic mathematical conventions.

8 Conclusion

In this work, we present Falcon-Arabic, a successful adaptation of Falcon3-7B to Arabic through vocabulary extension, multi-stage training, and exclusive use of native Arabic datasets. Our methodology involved extending Falcon3-7B tokenizer, implementing a gradual training recipe that preserves existing capabilities while incorporating diverse Arabic linguistic varieties. Post-training phases including SFT and DPO further enhanced instruction-following and cultural alignment.

The resulting Falcon-Arabic demonstrates that targeted adaptation with high-quality, native data can achieve exceptional performance, outperforming models two times its size while maintaining strong cultural relevance and linguistic authenticity. Our work provides valuable insights for effective language model adaptation strategies, showing that careful attention to tokenization, training design, and data authenticity can yield powerful models for underrepresented languages with limited computational resources. Future work will focus on improving the model on multiple areas including math, culture and RAG style of questions.

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A English Example

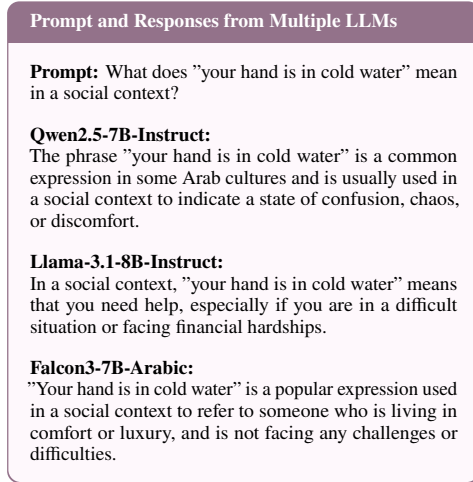


Figure 4: English translation of the example in Figure 1.

B Arabic Benchmarks

Throughout training, we assessed each checkpoint across training stages using Open Arabic LLM leaderboard benchmarks (El Filali et al., 2025). As shown in Figure 5, the majority of performance gains occurred during the initial stage when the model first encountered Arabic data, demonstrating the effectiveness of our tokenizer extension and layer initialization approach. Subsequent stages provided marginal improvements while primarily serving to expand the context window and prevent overfitting.

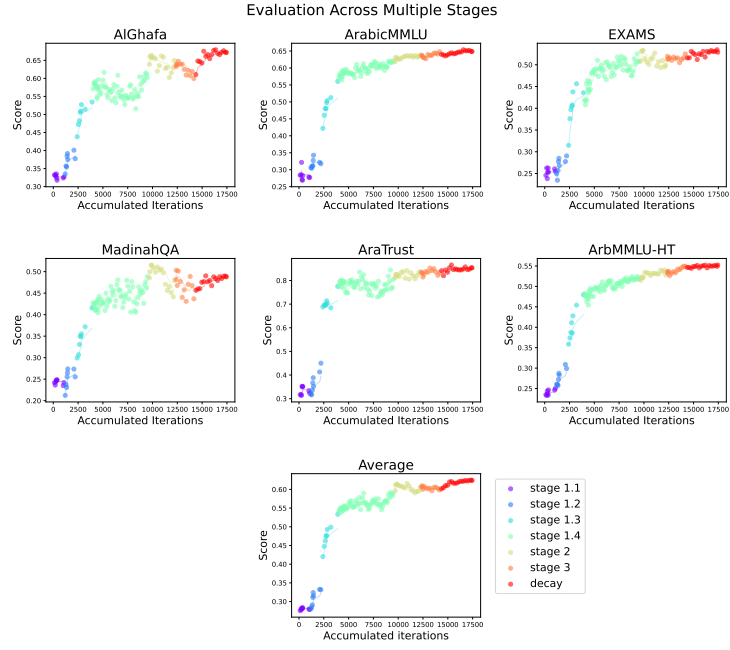


Figure 5: Scores evolution across multiple training stages of Falcon-Arabic on Arabic benchmarks.

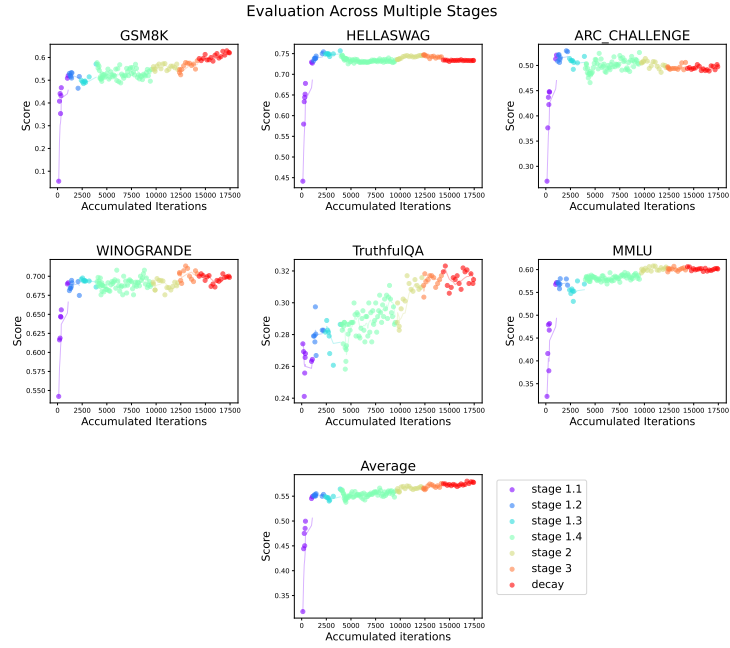


Figure 6: Scores evolution across multiple training stages of Falcon-Arabic on English benchmarks.